

# R1 Mock 1 — Classic SVNIT pattern

20 minutes · Ojas Srivastava · Intern – Technology

Do this first. Notes allowed once. Then run Mock 2 with no notes.

**From reports:** Sarthak Chauhan (SVNIT → GEP intern → PPO): intro → OOP pillars + polymorphism/inheritance with real examples → project deep dive → DSA.

*Interviewer tone: friendly, keeps pushing “give me a real example.”*

**How to use.** Set a 20-minute timer. Cover the green boxes. Answer out loud. Uncover only after you finish.

**Do NOT volunteer:** location, IFFCO (unless they open resume), MCP names, AirHelp, “I led the whole team.”

**Locked intro — say verbatim, then stop:**

Good afternoon sir, ma’am. I am Ojas Srivastava. Pre-final B.Tech AI, SVNIT Surat, CGPA 9.20. Coursework: OOP, DBMS, OS, CN, DAA. Google Solution Challenge 2026 Global rank 106 — LogiFlow, supply-chain routing. Vibe2Ship 2026 Global Top 20 — Community Hero, civic issue reporting PWA. Coding workflow: Cursor, MCP and plugins; spec and review every merge. Chairperson Nexus SVNIT, CSE+AI cell, manage and mentor 500+ students. I own the outcome. GEP works on AI-powered procurement and supply chain — operational data to a decision, owned till it ships.

## 1. Clock

|                    |          |                                                          |
|--------------------|----------|----------------------------------------------------------|
| <b>0:00–2:00</b>   | Intro    | Locked intro. Then <b>silence</b> .                      |
| <b>2:00–8:00</b>   | OOP      | Pillars + polymorphism / inheritance with real examples. |
| <b>8:00–14:00</b>  | LogiFlow | Architecture, your role, cache story.                    |
| <b>14:00–18:00</b> | DSA      | First unique character. Approach before code.            |
| <b>18:00–20:00</b> | Close    | One question for them.                                   |

## 2. Script

### 0:00 — Opening

**Interviewer.** *Good afternoon, Ojas. I have your resume. Tell me about yourself.*

**You say.** Locked intro. Stop.

### 2:00 — OOP block

**Interviewer.** *Explain the four pillars of OOP. Real-world examples — not textbook definitions.*

**You say.** Encapsulation — route service hides Redis; callers only see `getRoutes()`. Abstraction — steering wheel, no rack detail. Inheritance — Vehicle → Car/Truck. Polymorphism — `cost()` differs for railway vs road route.

**Interviewer.** *Go deeper on polymorphism and inheritance. Where did you use both?*

**You say.** Three LogiFlow pipelines share output shape. Ranker calls same interface on railway, comparator, hybrid — ranks time, cost, delay. Inheritance = reuse; polymorphism = one call, many behaviours.

**Interviewer.** *Compile-time or runtime polymorphism? One example each.*

**You say.** Overloading = compile-time — same name, different params. Overriding = runtime — subclass method through base reference. LogiFlow ranker on pipeline output is runtime.

**Interviewer.** *What is the purpose of OOP? Not “reusability.”*

**You say.** Model system as objects owning data and behaviour. Change stays local — swap railway pipeline without touching ranker. Reuse is a side effect.

### 8:00 — LogiFlow deep dive

**Interviewer.** *Walk me through LogiFlow. What problem and what was your role?*

**You say.** Ranks freight routes on time, cost, delay risk. Technical co-lead. 580+ corridors, 15k+ train-days. FastAPI, Next.js, Redis — 100–400 ms. pytest 100/100 business rules.

**Interviewer.** *Draw the flow — user query to ranked result.*

**You say.** Request → FastAPI → Redis check (cache-aside) → on miss: pipeline fetch/transform → ranker scores → cache + return → Next.js. SQL is source of truth.

**Interviewer.** *Why Redis? Stale cache? Redis down?*

**You say.** In-memory, sub-ms reads. TTL + invalidate on write. Redis down — recompute from SQL; slower but correct. Cache never decides business truth.

**Interviewer.** *What difficulty did you face?*

**You say.** Latency. First version recomputed every request — seconds. Cache-aside cut it to 100–400 ms.

### 14:00 — DSA

**Interviewer.** *Find the first non-repeating character. Talk before you write.*

**You say.** Two passes: count in hashmap, scan left-to-right for count 1. “swiss” → w.  $O(n)$  time,  $O(1)$  space fixed alphabet.

**Code sketch.** Pass 1 count; pass 2 return first with count 1. Empty string → sentinel.

### 18:00 — Close

**Interviewer.** *Any questions for us?*

**You say.** “For technology interns, do early projects lean more toward SMART procurement or NEXXE supply chain?”

## 3. Score this mock

Rate each area /10. Pass bar: 7+ average.

|                 |                                                            |       |
|-----------------|------------------------------------------------------------|-------|
| <b>Intro</b>    | Under 60s? Stopped? No location / IFFCO / MCP names?       | _____ |
| <b>OOP</b>      | LogiFlow example in first two sentences?                   | _____ |
| <b>LogiFlow</b> | 580+, 15k+, 100–400 ms, 100/100, co-lead not “whole team”? | _____ |
| <b>DSA</b>      | Approach + complexity before code? swiss → w?              | _____ |
| <b>Close</b>    | One smart question, not stipend?                           | _____ |

**Pass bar.** OOP answers had LogiFlow in the first two sentences. DSA clean in one pass with complexity stated.

Next: GEP\_R1\_Mock\_2.pdf — no notes.